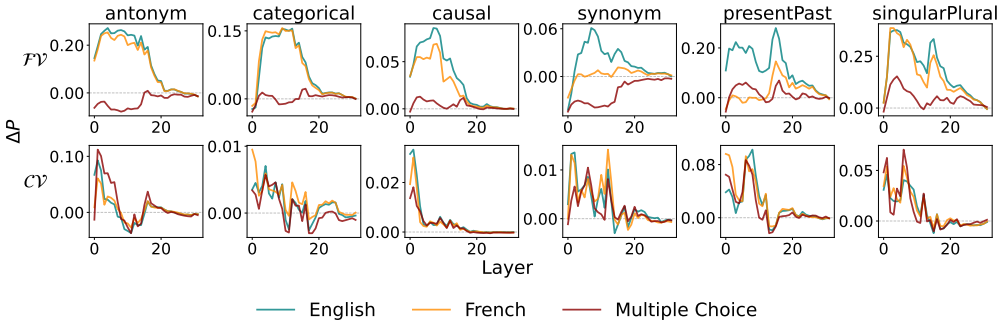
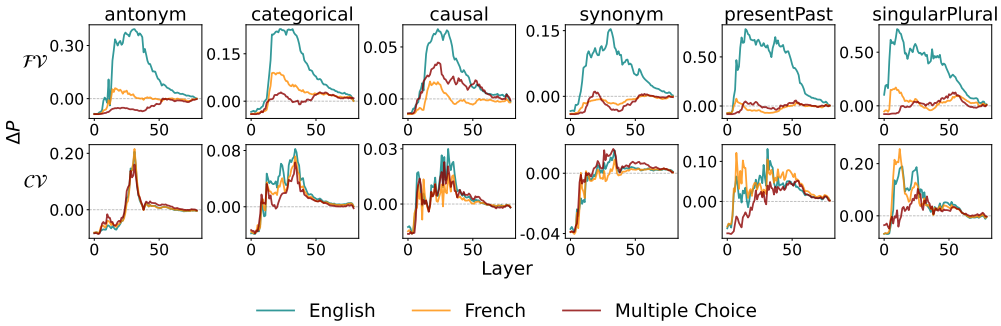
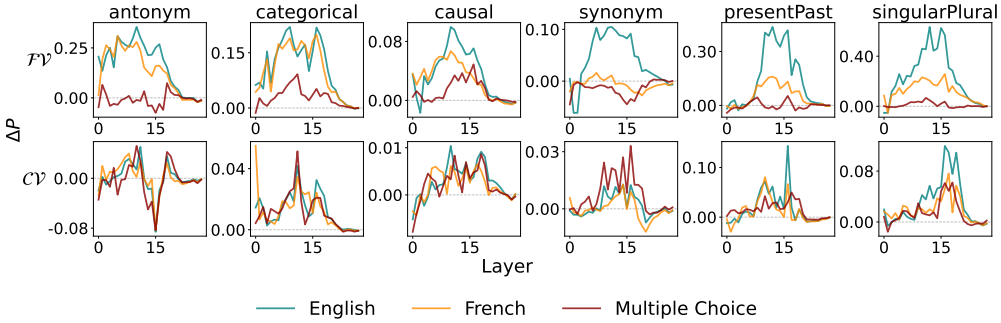
H STEERING RESULTS



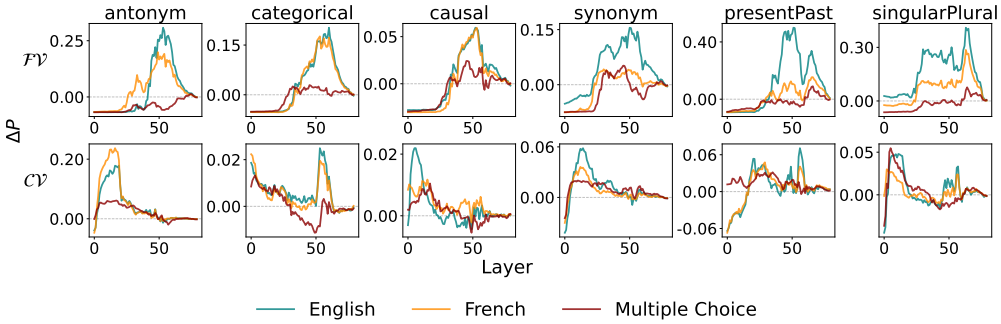
(a) Llama 3.1 8B



(b) Llama 3.1 70B



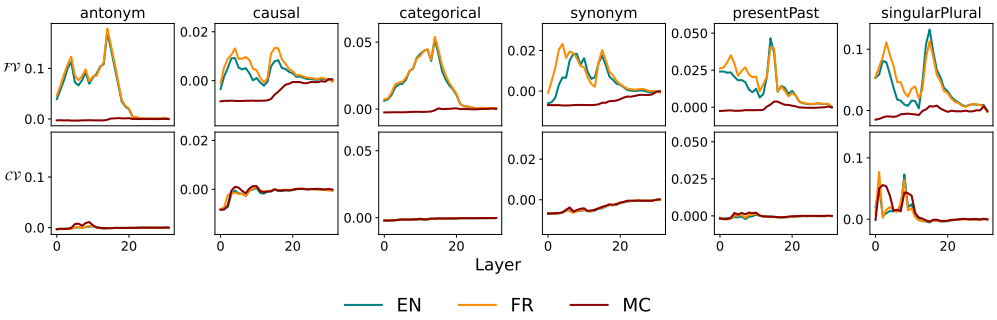
(c) Qwen 2.5 7B



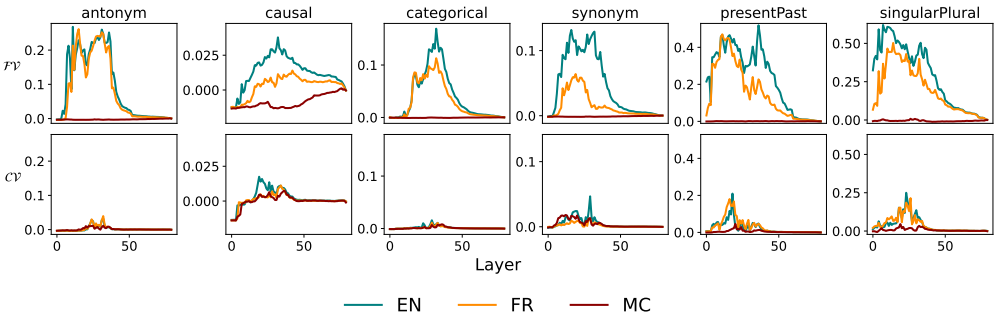
(d) Qwen 2.5 72B

Figure 16: Steering effect across layers and all concepts for different models.

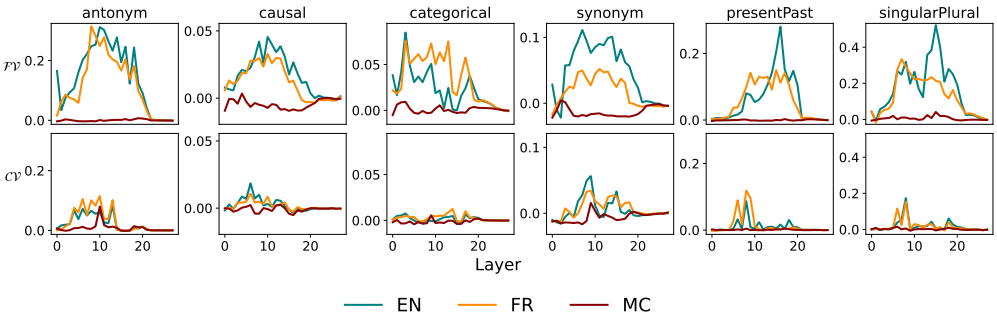
I 0-SHOT STEERING RESULTS



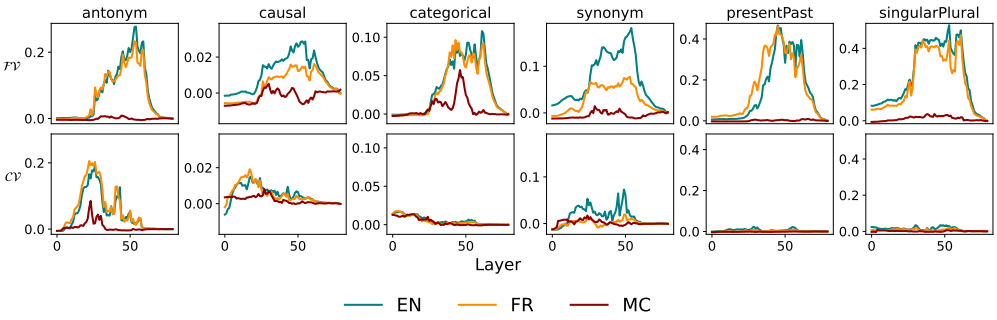
(a) Llama 3.1 8B



(b) Llama 3.1 70B



(c) Qwen 2.5 7B



(d) Qwen 2.5 72B

Figure 17: 0-shot steering effect across layers and all concepts for different models.

J QWEN 2.5 72B OUTLIER ANALYSIS

We identified anomalous CIE values for Qwen 2.5 72B in the Categorical concept across French open-ended and multiple-choice formats. As shown in Figure 18, these conditions exhibit unusually high CIE values with a bimodal distribution that deviates from the expected pattern. We excluded these two datasets from the final AIE calculations. This exclusion has minimal impact on our results: the top-5 head rankings remain identical (100% overlap), confirming that our main findings are robust to this methodological decision.

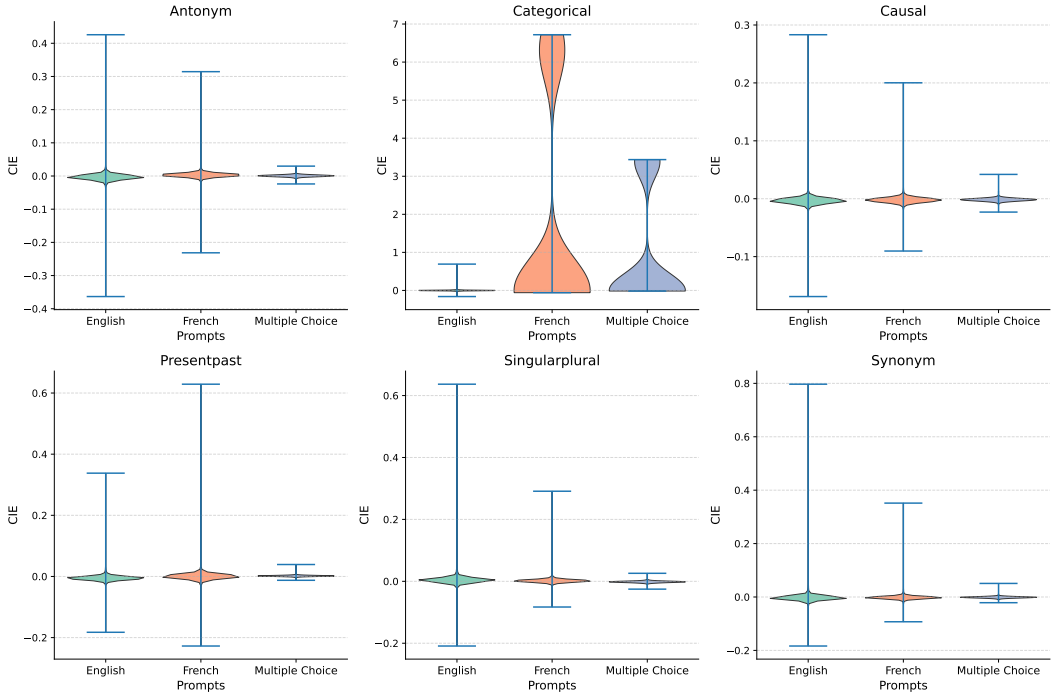


Figure 18: Violin plots of CIE for different concepts and prompts.